

Software Engineering Lab 3

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Section: B

Assigned Topic Number: 6

Assigned Topic: Alumni Mentorship and Mock Interview Platform

Architecture Selection: I chose **Layered Architecture** for the Alumni Mentorship & Mock Interview Platform.

Reason 1:

Clear separation of concerns: The platform has distinct responsibilities (UI, matching and booking logic, data storage). Layering keeps these apart, so the matching algorithm or booking rules can change without touching the screens or the database.

Reason 2:

Centralised business rules: Rules such as preventing double-booking of mentor slots (FR-001) and the 10-minute scorecard visibility window (FR-004) live in one Business Layer. This keeps them consistent and easy to test.

Security Advantage: All requests pass through the Business Layer and the Authentication Service before reaching the database, so the data layer is never exposed directly. This supports NFR-001 (privacy and soft-deletion) and NFR-002 (alumni contact info visible only to authenticated students).

Performance Benefit: The Business Layer can cache mentor availability and matching results, so students see updated slots within the 5-minute target of FR-002 without querying the database on every request.